

CSA6501
Generative AI
Lab Programs

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1. Aim:

To install Matplotlib and display a simple line graph.

Algorithm

1. Import the Matplotlib library.
2. Create X and Y values.
3. Plot the line graph.
4. Add title and labels.
5. Display the graph.

Code:

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 6, 8, 10]
```

```
plt.plot(x, y)
```

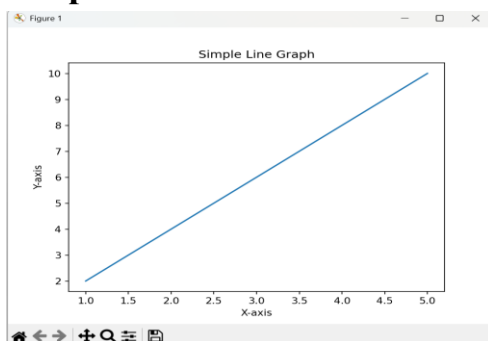
```
plt.title("Simple Line Graph")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
plt.show()
```

Output:



Result:

Thus, a simple line graph was created successfully using Matplotlib.

2. Aim:

To verify the installed version of Scikit-learn.

Algorithm

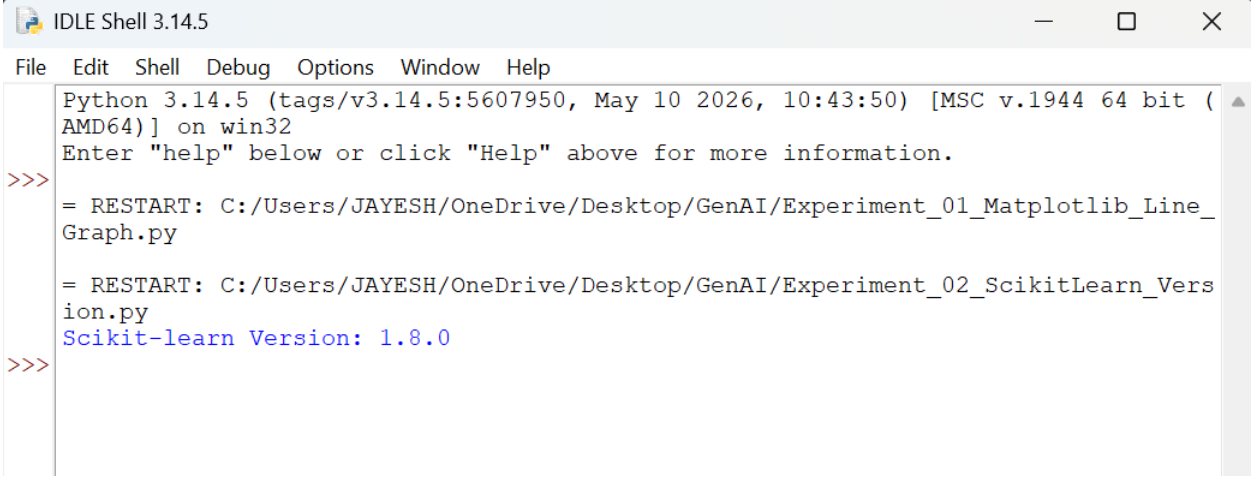
1. Import Scikit-learn.
2. Print the version.
3. Display the output.

Code:

```
import sklearn
```

```
print("Scikit-learn Version:", sklearn.__version__)
```

Output:



```
IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/JAYESH/OneDrive/Desktop/GenAI/Experiment_01_Matplotlib_Line_Graph.py
= RESTART: C:/Users/JAYESH/OneDrive/Desktop/GenAI/Experiment_02_ScikitLearn_Version.py
Scikit-learn Version: 1.8.0
>>>
```

Result:

Thus, the installed version of Scikit-learn was displayed successfully.

3. Aim

To create a NumPy array containing the first 20 natural numbers.

Algorithm

1. Import NumPy.
2. Create an array using arange().
3. Display the array.

Code

```
import numpy as np
```

```
arr = np.arange(1, 21)
```

```
print(arr)
```

Output:

```
>>> = RESTART: C:/Users/JAYESH/OneDrive/Desktop/GenAI/Experiment_03_NumPy_Array.py =
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
>>> |
```

Result:

Thus, the NumPy array containing the first 20 natural numbers was created successfully.

4. Aim

To create a Pandas DataFrame containing student names and grades.

Algorithm

1. Import Pandas.
2. Create a dictionary.
3. Convert it into a DataFrame.
4. Display the DataFrame.

Code

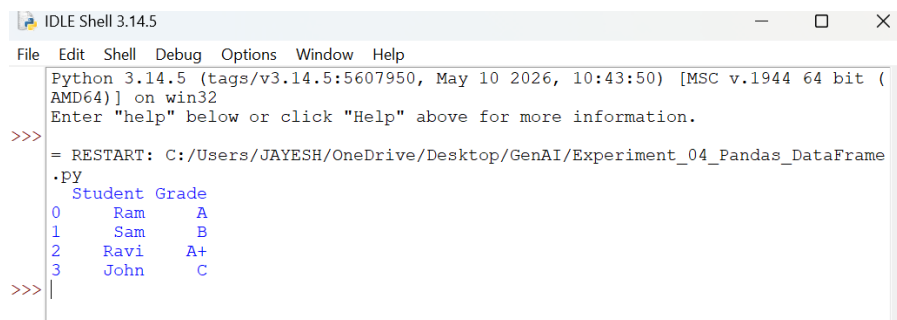
```
import pandas as pd

data = {
    "Student": ["Ram", "Sam", "Ravi", "John"],
    "Grade": ["A", "B", "A+", "C"]
}

df = pd.DataFrame(data)
```

```
print(df)
```

Output:



```
IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/JAYESH/OneDrive/Desktop/GenAI/Experiment_04_Pandas_DataFrame
.py
      Student Grade
0         Ram    A
1         Sam    B
2         Ravi  A+
3         John    C
>>> |
```

Result:

Thus, the Pandas DataFrame was created successfully.

5. Aim: To display the installed versions of commonly used AI libraries.

Algorithm

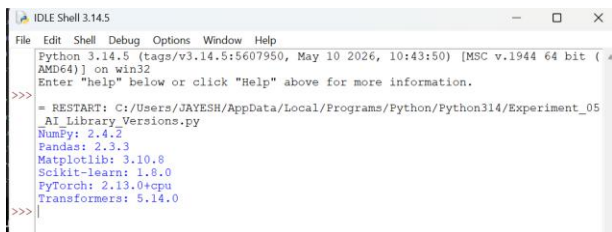
1. Import all required libraries.
2. Print their versions.
3. Display the output.

Code

```
import numpy
import pandas
import matplotlib
import sklearn
import torch
import transformers

print("NumPy:", numpy.__version__)
print("Pandas:", pandas.__version__)
print("Matplotlib:", matplotlib.__version__)
print("Scikit-learn:", sklearn.__version__)
print("PyTorch:", torch.__version__)
print("Transformers:", transformers.__version__)
```

Output:



```
IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/JAYESH/AppData/Local/Programs/Python/Python314/Experiment_05
AI_Library_Versions.py
NumPy: 2.4.2
Pandas: 2.3.3
Matplotlib: 3.10.8
Scikit-learn: 1.8.0
PyTorch: 2.13.0+cpu
Transformers: 5.14.0
>>>
```

Result:

Thus, the installed versions of the AI libraries were displayed successfully.

6. Aim

To create tensors using PyTorch and perform basic mathematical operations.

Algorithm

1. Import the PyTorch library.
2. Create two tensors.
3. Perform addition and multiplication.
4. Display the results.

Code

```
import torch

a = torch.tensor([1, 2, 3])

b = torch.tensor([4, 5, 6])

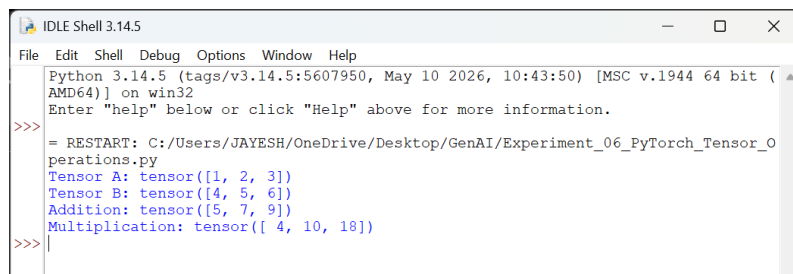
print("Tensor A:", a)

print("Tensor B:", b)

print("Addition:", a + b)

print("Multiplication:", a * b)
```

Output:



```
IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/JAYESH/OneDrive/Desktop/GenAI/Experiment_06_PyTorch_Tensor_Operations.py
Tensor A: tensor([1, 2, 3])
Tensor B: tensor([4, 5, 6])
Addition: tensor([5, 7, 9])
Multiplication: tensor([ 4, 10, 18])
>>> |
```

Result:

Thus, tensors were created successfully and basic mathematical operations were performed using PyTorch.

7. **Aim:** To load a pre-trained Transformer model using Hugging Face and perform sentiment analysis.

Algorithm

1. Import the Hugging Face pipeline.
2. Load the sentiment analysis model.
3. Provide the input sentence.
4. Display the sentiment result.

Code

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

text = "I love Artificial Intelligence."

result = classifier(text)

print(result)
```

Output:

```
Warning (from warnings module):
  File "C:\Users\JAYESH\AppData\Local\Programs\Python\Python314\Lib\site-packages\huggingface_hub\file_download.py", line 139
    warnings.warn(message)
UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\JAYESH\AppData\Local\Programs\Python\Python314\Lib\site-packages\huggingface_hub\models--distilbert--distilbert-base-uncased-finetuned-sst-2-english. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
Loading weights: 0%|          | 0/104 [00:00<?, ?it/s]Loading weights: 100%|██████████| 104/104 [00:00<00:00, 20399.74it/s]
tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<?, ?B/s]tokenizer_config.json: 100%|██████████| 48.0/48.0 [00:00<00:00, 7.63kB/s]
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]vocab.txt: 100%|██████████| 232k/232k [00:01<00:00, 191kB/s]vocab.txt: 100%|██████████| 232k/232k [00:01<00:00, 182kB/s]
[{'label': 'POSITIVE', 'score': 0.9994176626205444}]
```

Result:

Thus, sentiment analysis was performed successfully using the Hugging Face pre-trained model.

8. **Aim:** To tokenize a sentence using the BERT tokenizer.

Algorithm

1. Import the BERT tokenizer.
2. Load the pre-trained tokenizer.
3. Enter a sentence.
4. Tokenize the sentence.
5. Display the tokens.

Code:

```
from transformers import BertTokenizer

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

text = "Artificial Intelligence is amazing."

tokens = tokenizer.tokenize(text)

print("Tokens:", tokens)
```

Output

```
===== RESTART: C:/Users/JAYESH/OneDrive/Desktop/GenAI/Experiment_06_BERT_tokenization.py =====
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<?, ?B/s]tokenizer_config.json: 100%|██████████| 48.0/48.0 [00:00<00:00, 14.0kB/s]

Warning (from warnings module):
  File "C:/Users/JAYESH/AppData/Local/Programs/Python/Python314/Lib/site-packages/huggingface_hub/file_download.py", line 139
    warnings.warn(message)
UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\JAYESH\cache\huggingface\hub\models--bert-base-uncased. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]vocab.txt: 100%|██████████| 232k/232k [00:00<00:00, 375kB/s]vocab.txt: 100%|██████████| 232k/232k [00:00<00:00, 359kB/s]
tokenizer.json: 0%|          | 0.00/466k [00:00<?, ?B/s]tokenizer.json: 100%|██████████| 466k/466k [00:01<00:00, 453kB/s]tokenizer.json: 100%|██████████| 466k/466k [00:01<00:00, 447kB/s]
Tokens: ['artificial', 'intelligence', 'is', 'amazing', '.']
```

Result:

Thus, the sentence was successfully tokenized using the BERT tokenizer.

